Finite Self-Models and the Open Explanatory Gap: A Conditional Arithmetic Analysis

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July 2026

Abstract

Paper G offers a programme-level diagnosis of possible self-reference limits, while Paper II retains a leading small-ball stress-tensor extraction with an $O(R^{d+3})$ remainder. Neither result supplies a fixed-point no-go theorem, a physical Fisher spectrum, or a self-description tower. This paper tests a proposed quantitative bridge by defining a local Araki-relative-entropy reconstruction loss for a stipulated finite statistical bottleneck.

The former local minimax drift-floor theorem is withdrawn. A finite Fisher ceiling does not by itself imply a positive reconstruction loss: on a short modular orbit, an admissible two-outcome channel can preserve the orbit coordinate exactly and achieve zero loss. A future positive risk theorem would therefore need a typed observation model that excludes such lossless-orbit constructions and relates estimator risk to the reconstruction loss.

The surviving formal result is a conditional arithmetic inversion. If a stipulated positive sequence satisfies

$$\log I_k = -\alpha k \log k + O(k)$$

and the available budget grows at polynomial scale, then its maximal admissible index obeys

$$K(n) = O\left(\frac{\log n}{\log\log n}\right).$$

This lemma proves neither a physical coefficient/Fisher spectrum nor a self-model resource bound. The explanatory gap therefore remains open: the paper identifies conditions required for a future quantitative rebuild but does not establish a finite self-description horizon.

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1 Introduction

Paper G[3] motivates the possibility that self-reference limits an accessibility-based description. Its regress and collapse scenarios are philosophical diagnostics, not a fixed-point no-go theorem or an exhaustive dichotomy. Paper III uses that discussion only as motivation.

Paper II[2] moved the discussion to a more concrete level. In the higher-dimensional CHM ball framework, Paper II retains the leading stress-tensor extraction at order R^{d+1}, with a remainder $O(R^{d+3})$. It does not construct independent coefficient directions, a Fisher map, or a self-description tower from that remainder.

The present paper defines a local Araki-relative-entropy reconstruction loss for a stipulated finite statistical bottleneck, but withdraws the claimed universal drift floor. Estimation risk in a specified noisy experiment cannot be identified with reconstruction loss for a channel that may preserve the orbit coordinate exactly. The surviving formal result is instead a conditional arithmetic inversion for a stipulated positive sequence and budget. The surrounding programme draws on:

1. the BKM/QFIM structure already used in Paper I[1];
2. the self-referential Fisher and ego-rigidity formalism of T-DOME Paper III[4];
3. the leading CHM expansion and remainder order recorded in Paper II[2].

The arithmetic statement assumes, rather than derives, a positive sequence I_k with

$$\log I_k = -\alpha k \log k + O(k),$$

and a polynomial-scale budget. Under those stipulations its maximal index obeys

$$K(n) = O\left(\frac{\log n}{\log\log n}\right).$$

This proves neither a physical Fisher spectrum nor a self-model budget.

The trilogy-level interpretation also depends on Paper I's supplied shell interface (Amb), its dynamical covariance axiom (C), and the scale-faithfulness axiom (SFa). Modular flow supplies algebraic time only; a phenomenal-time reading is conditional and interpretive. T-DOME Paper III supplies an ego-rigidity prior model, not a theorem that turns it into a reconstruction floor.

The paper is organized as follows. Section 2 defines finite observers, admissible self-models, and the self-description deficit. Section 3 records why the former drift-floor theorem is withdrawn. Section 4 proves the conditional arithmetic inversion. Section 5 retains only the exact CHM moment coefficient and explains why it does not supply a Fisher spectrum. Section 6 explains the trilogy-level significance of the result, and Section 7 states the remaining limitations and open problems.

2 Finite Observers and the Self-Description Deficit

2.1 Finite Observers

Motivated by Paper G's architectural discussion, we first specify what counts as a finite statistical self-model. This definition does not inherit a no-go theorem from Paper G.

Definition 2.1 (Finite observer)**.** A finite observer of complexity n is a triple

$$(\mathcal{A}_{\text{obs}}, \omega_{\text{obs}}, \mathfrak{M}_n),$$

where:

1. $\mathcal{A}_{\text{obs}} \subset \mathcal{A}_c$ is the observer's accessible subalgebra;
2. $\omega_{\text{obs}} = \omega|_{\mathcal{A}_{\text{obs}}}$ is the restricted state;
3. $\mathfrak{M}_n$ is an n-dimensional statistical model used internally to represent the observer's own accessible state.

The complexity parameter n measures effective statistical degrees of freedom, not arbitrary description length. This is deliberate. Paper III is not about Kolmogorov complexity in full generality. It is about the finite number of statistically usable directions available to an observer trying to model itself.

2.2 Modular Fisher Sensitivity

The Tomita–Takesaki modular automorphism group σ_t^ω of a faithful normal state ω on a von Neumann algebra satisfies $\omega \circ \sigma_t^\omega = \omega$ for all t. The orbit is therefore trivial at the level of states, and the associated tangent vector vanishes. Since the modular orbit itself carries no state-level Fisher information, we replace it with the intrinsic accessible-unitary perturbation family as the operational proxy for modular sensitivity.

Definition 2.2 (Modular Fisher sensitivity)**.** Fix a self-adjoint element $h \in \mathcal{A}_{\text{obs}}$ and define the perturbed family

$$\omega_s^{(h)} := \omega_{\text{obs}} \circ \text{Ad}(e^{ish}), \qquad s \in (-\epsilon, \epsilon).$$

The modular Fisher sensitivity of ω_{obs} along h is

$$\mathcal{I}_F(\omega_{\text{obs}}, h) := \mathbf{Q}_{\omega_{\text{obs}}}\left(\left.\frac{d\omega_s^{(h)}}{ds}\right|_{s=0}, \left.\frac{d\omega_s^{(h)}}{ds}\right|_{s=0} \right).$$

The modular Fisher sensitivity of the observer is

$$\mathcal{I}_F(\omega_{\text{obs}}) := \sup\big\{ \mathcal{I}_F(\omega_{\text{obs}}, h) : \ h = h^* \in \mathcal{A}_{\text{obs}}, \ \|h\| \le 1 \big\}.$$

The supremum is taken over unit-norm self-adjoint perturbations in the observer's own accessible algebra. It is finite whenever the BKM form is bounded on the unit ball of $\mathcal{A}_{\text{obs}}$-tangent directions, which holds in the local regime used throughout this paper. The quantity $\mathcal{I}_F(\omega_{\text{obs}})$ measures the observer's maximal intrinsic sensitivity to unitary perturbations of its own accessible state. It is strictly positive whenever ω_{obs} is not a trace state on $\mathcal{A}_{\text{obs}}$.

Throughout the paper, the term *local modular drift family* refers to the one-parameter family $\{\omega_s^{(h)}\}$ for a fixed perturbation h. When the supremum in Definition 2.2 is attained, we take h to be a maximizer; when it is not, we use a standard η-optimization: for each $\eta > 0$, choose $h_\eta = h_\eta^* \in \mathcal{A}_{\text{obs}}$ with $\|h_\eta\| \le 1$ and $\mathcal{I}_F(\omega_{\text{obs}}, h_\eta) \ge \mathcal{I}_F(\omega_{\text{obs}}) - \eta$, carry out the proof for h_η, and let $\eta \downarrow 0$. Despite the name, this family is not the Tomita–Takesaki modular orbit (which is state-preserving) but the accessible-unitary perturbation family that serves as its operational proxy at the level of Fisher information. This is the family along which the observer's self-description is hardest to compress.

2.3 Admissible Self-Models

The theorem chain does not require a literal embedding of the self-model into a matrix algebra. That stronger route would force us to solve a much harder operator-algebraic problem than the present paper needs. What matters here is the existence of a finite statistical bottleneck.

Definition 2.3 (Admissible self-model)**.** An admissible n-dimensional self-model is a pair

$$(\Pi_n, \Psi_n),$$

where:

1. $\Pi_n : \mathcal{S}(\mathcal{A}_{\text{obs}}) \to \mathfrak{M}_n$ is the state-space map induced by a normal unital completely positive map from a finite-dimensional algebra into $\mathcal{A}_{\text{obs}}$, so that Petz monotonicity of relative entropy applies;
2. $\Psi_n : \mathfrak{M}_n \to \mathcal{S}(\mathcal{A}_{\text{obs}})$ is a recovery map;
3. Π_n factors through the observer's own accessible state space, so no external "God's-eye" parameter is introduced;
4. near the reference state ω_{obs}, the Fisher matrix on $\mathfrak{M}_n$ can be whitened:
$$\mathbf{I}_{\omega}^{(n)} = \text{diag}(\lambda_1, \ldots, \lambda_n), \qquad \lambda_a \leq \mathcal{I}_F(\omega_{\text{obs}}).$$

The last condition is the substantive standing assumption of the theorem, not a mere coordinate normalization. It says that the compressed model has only n effective Fisher directions, and none of them is sharper than the observer's own modular sensitivity scale. This is the exact place where the theorem absorbs finite-dimensionality.

2.4 Self-Description Deficit

The correct loss function is not a metric but a divergence. In a Type III setting there is no preferred trace-class density-matrix picture, while Araki relative entropy is native to the operator algebraic formulation [5].

Definition 2.4 (Self-description deficit)**.** Define the BKM ball of radius δ around ω_{obs} by

$$\mathcal{B}_\delta(\omega_{\text{obs}}) := \left\{\omega \in \mathcal{S}(\mathcal{A}_{\text{obs}}) : \ \mathbf{Q}_{\omega_{\text{obs}}}(\omega - \omega_{\text{obs}},\, \omega - \omega_{\text{obs}}) < \delta^2\right\}.$$

Fix once and for all a $\delta > 0$ small enough that Araki relative entropy admits its quadratic expansion on $\mathcal{B}_\delta(\omega_{\text{obs}})$, and large enough that the ego-rigidity prior π_{ego} of T-DOME Paper III[4] can be supported on a segment of the local modular drift family inside $\mathcal{B}_\delta(\omega_{\text{obs}})$. The self-description deficit of complexity n is

$$\varepsilon(n) := \inf_{(\Pi_n, \Psi_n)} \sup_{\omega \in \mathcal{B}_\delta(\omega_{\text{obs}})} S(\omega \parallel \Psi_n \circ \Pi_n(\omega))\,,$$

where the infimum is taken over all admissible n-dimensional self-models.

Near a fixed state, Araki relative entropy has the familiar BKM/QFIM quadratic expansion

$$S(\omega \| \omega + \delta\omega) = \frac{1}{2}\mathbf{Q}_\omega(\delta\omega, \delta\omega) + o(\|\delta\omega\|^2),$$

which is the local information-geometric structure already used in Paper I[1] and in Petz's monotone-metric framework [7]. The minimax form is essential: it rules out trivial constant recovery maps that memorize the reference state while failing arbitrarily nearby. Thus $\varepsilon(n)$ measures optimal local uniform reconstruction loss in the intrinsic Type III geometry of states.

3 The Fisher Ceiling for Admissible Self-Models

3.1 Petz Monotonicity and Finite Fisher Budget

The first ingredient is standard.

Lemma 3.1 (Petz monotonicity)**.** *For any normal coarse-graining* Π_n*,*

$$S(\omega\|\sigma) \geq S(\Pi_n(\omega)\|\Pi_n(\sigma)).$$

Proof. This is the monotonicity of relative entropy under coarse-graining [6]. The present paper uses it only in the local regime, where it implies that compression cannot increase BKM/Fisher information. □

Whitening then converts finite statistical dimension directly into two distinct ceilings: one for the specific modular drift channel, and one for the total trace budget across all retained directions.

Proposition 3.2 (Directional and total Fisher ceilings)**.** *Let* (Π_n, Ψ_n) *be an admissible* n*-dimensional self-model. Then:*

1. *along the local modular drift family through* ω_{obs}*, the Fisher information retained after compression satisfies*
$$\mathcal{I}^{(n)}_{\text{drift}} \leq \mathcal{I}_F(\omega_{\text{obs}});$$
2. *the total retained data Fisher information satisfies*
$$\mathcal{I}^{(n)}_{\text{data}} := \operatorname{tr} \mathbf{I}^{(n)}_\omega \leq n\, \mathcal{I}_F(\omega_{\text{obs}}).$$

Proof. (1) Apply Lemma 3.1 to the one-parameter modular drift family. In the local quadratic regime, monotonicity of relative entropy implies monotonicity of the associated BKM/Fisher metric, so compression cannot increase Fisher information for that specific scalar parameter.

(2) In Fisher-whitened coordinates, the compressed Fisher matrix is

$$\mathbf{I}^{(n)}_\omega = \operatorname{diag}(\lambda_1, \ldots, \lambda_n), \qquad \lambda_a \leq \mathcal{I}_F(\omega_{\text{obs}}).$$

Summing over the n effective directions gives

$$\operatorname{tr} \mathbf{I}^{(n)}_\omega = \sum_{a=1}^{n} \lambda_a \leq n\, \mathcal{I}_F(\omega_{\text{obs}}).$$

□

3.2 Withdrawal of the Drift-Floor Claim

The former local minimax drift-floor theorem and its corollary are withdrawn. The argument applied a van Trees estimation-risk bound to a compression/recovery loss without specifying a noisy observation model that prevents the orbit coordinate from being retained.

The obstruction can be implemented by an internal finite-dimensional channel. Let $s \mapsto \omega_s$ be a nonconstant smooth local drift orbit. Choose an effect $0 \leq M \leq \mathbf{1}$ in $\mathcal{A}_{\text{obs}}$ with $\frac{d}{ds}\omega_s(M)|_{s=0} \neq 0$. The normal unital completely positive map

$$\Phi : \mathbb{C}^2 \longrightarrow \mathcal{A}_{\text{obs}}, \qquad \Phi(a, b) = aM + b(\mathbf{1} - M),$$

induces the binary statistical coordinate

$$\Pi_1(\omega_s) = \big(p(s), 1 - p(s)\big), \qquad p(s) := \omega_s(M).$$

By the inverse-function theorem, after restricting to a sufficiently short orbit segment, p is injective. Define the recovery map on its image, and extend it arbitrarily off that image, by

$$\Psi_1\big(p(s), 1 - p(s)\big) = \omega_s.$$

This code uses an observable in the accessible algebra rather than an external parameter, and it factors through a finite-dimensional commutative algebra as required by Definition 2.3. Then

$$S(\omega_s \| \Psi_1 \circ \Pi_1(\omega_s)) = 0$$

for every s on the encoded segment. A positive Bayes estimation risk can arise only after a likelihood or sampling channel restricts access to s; it is not a lower bound on this lossless orbit reconstruction. The previous proof therefore confused estimation risk with reconstruction loss. Petz monotonicity and the stipulated Fisher ceilings above remain bookkeeping facts within their stated model, but they imply no nonzero reconstruction floor.

4 A Conditional Arithmetic Inversion

Paper II[2] supplies a leading CHM coefficient and an $O(R^{d+3})$ remainder, not an infinite family of normalized Fisher directions. We therefore separate the arithmetic statement from any physical interpretation.

Let $\{I_k\}_{k\geq 1}$ be a *stipulated* positive sequence and let $B(n) \geq 1$ be a *stipulated* budget. Define

$$K(n) := \max\left\{ K \geq 1 : \sum_{j=1}^{K} I_j^{-1} \leq B(n) \right\},$$

with $K(n) = 0$ if the set is empty. No claim is made here that the I_k are Fisher informations of physical coefficient directions or that $B(n)$ is a self-model resource.

Lemma 4.1 (Conditional arithmetic inversion)**.** *Suppose*

$$\log I_k = -\alpha k \log k + O(k), \qquad \alpha > 0,$$

and $\log B(n) = O(\log n)$. *Then*

$$K(n) = O\left(\frac{\log n}{\log\log n}\right).$$

Proof. If $K = K(n) \geq 1$, the final nonnegative summand satisfies $I_K^{-1} \leq B(n)$. Hence

$$\alpha K \log K + O(K) = -\log I_K \leq \log B(n) = O(\log n).$$

For sufficiently large K, the $K \log K$ term dominates the linear remainder, so inversion gives $K = O(\log n / \log\log n)$. Bounded K obeys the same asymptotic bound. □

This lemma is exactly the surviving claim. It establishes neither a self-model budget nor a physical Fisher spectrum. In particular, rescaling coefficient coordinates changes any candidate I_k, so a physical application requires a separately justified normalization, likelihood, and Fisher map.

5 Exact CHM Moment Without a Fisher-Spectrum Claim

The following exact integral is retained as an algebraic check. It does not construct the coefficient directions or Fisher map absent from Paper II.

5.1 Exact d=1 Signal Factor

In $d = 1$, the CHM modular Hamiltonian is

$$K_B = \frac{\pi}{R} \int_{-R}^{R} (R^2 - x^2)\, T_{00}(x)\, dx.$$

Expanding $T_{00}(x)$ around the ball center gives

$$T_{00}(x) = \sum_{k \geq 0} \frac{x^{2k}}{(2k)!} \partial_x^{2k} T_{00}(0) + \cdots .$$

Therefore the coefficient multiplying the $2k$-th derivative is

$$c_k^{(d=1)} = \frac{\pi}{R} \cdot \frac{1}{(2k)!} \int_{-R}^{R} (R^2 - x^2) x^{2k}\, dx \cdot R^{-(2k+2)}.$$

The radial moment is exact:

$$\int_{-R}^{R} (R^2 - x^2) x^{2k}\, dx = \frac{4R^{2k+3}}{(2k+1)(2k+3)}.$$

Hence

$$c_k^{(d=1)} = \frac{4\pi}{(2k+1)(2k+3)(2k)!},$$

which shows exact factorial suppression:

$$c_k^{(d=1)} \sim \frac{\text{const}}{(2k)!\, k^2}.$$

The factorial suppression of $c_k^{(d=1)}$ is normalization-dependent signal information. It does not by itself determine a fluctuation kernel, a renormalized Fisher information I_k, or a cumulative inverse-information budget. Consequently this calculation does not instantiate Lemma 4.1; it only shows what one factor in a future, fully specified model would look like.

6 Trilogy Implications

6.1 Paper G as Motivation Only

Paper G[3] supplies philosophical motivation for studying self-description, but no formal bridge maps its syntax-level scenarios to the reconstruction loss defined here. The withdrawn drift-floor argument does not quantify Paper G, and this paper asserts no generic fixed-point obstruction.

6.2 Relocating the Gap in Paper II Terms

Paper II[2] supplies a leading CHM stress-tensor term and an $O(R^{d+3})$ remainder. It does not supply normalized coefficient directions or a Fisher map. Paper III therefore does not reinterpret the remainder as a physical control tower. Lemma 4.1 applies only after a positive sequence and budget are stipulated independently.

6.3 Trilogy-Level Interpretation

The trilogy-level reading is conditional on Paper I's shell interface (Amb), dynamical axiom (C), and scale-faithfulness assumption (SFa), as well as Paper II's restricted CHM regime. These inputs supply algebraic modular time and conditional geometric statements, not a phenomenal-time identification. The surviving arithmetic lemma does not locate or quantify an explanatory gap.

7 Limitations and Open Problems

The withdrawals identify four concrete tasks for any future rebuild.

1. **A typed observation model.** A future risk theorem must specify a likelihood or sampling channel that prevents lossless orbit coding, then distinguish estimator risk from state reconstruction loss.

2. **Normalized coefficient directions.** A physical use of Lemma 4.1 requires independently constructed coefficient perturbations, a renormalized kernel, and a normalization under which the Fisher sequence is invariantly defined.

3. **A Paper II bridge.** The remainder order alone is insufficient. One must construct a map from small-ball coefficient data to the statistical model before any resource comparison is meaningful.

4. **Ego-rigidity interface.** The T-DOME prior may enter a specified Bayesian estimation problem, but it supplies no reconstruction floor by itself. Its operational channel and boundary conditions must be given before van Trees can be used.

8 Conclusion

Paper III tests a proposed quantitative route to the explanatory gap. Paper G supplies programme-level motivation, while Paper II supplies a leading stress-tensor extraction and a controlled remainder order; it does not identify that remainder as a physical self-description carrier. The present paper defines a finite-self-model reconstruction loss and records why the former universal drift-floor inference fails.

A finite Fisher ceiling alone does not imply a nonzero local minimax reconstruction loss. The lossless-orbit construction shows that a finite channel can preserve the relevant modular-orbit coordinate exactly on a short segment. A positive lower bound may still arise under an additional typed observation model that excludes such encodings and connects estimation risk to reconstruction loss, but a precise characterization of those conditions remains open.

Separately, the conditional arithmetic lemma shows that a stipulated positive sequence with $\log I_k = -\alpha k \log k + O(k)$ has maximal admissible index $K(n) = O(\log n / \log\log n)$ under a

polynomial-scale budget. It does not establish a physical Fisher spectrum, a self-model budget, or a finite self-description horizon. The explanatory gap is therefore not eliminated in general, but it is not quantitatively located by the present results.

A Trilogy Key Premises Summary

The following list records the external inputs and their restricted roles. Paper III is not self-contained with respect to these trilogy interfaces.

1. **Paper I algebraic interface** (Paper I[1]). The Araki relative entropy of two nearby faithful normal states on a Type III von Neumann algebra admits a local quadratic expansion $S(\omega\|\omega+\delta\omega) = \frac{1}{2}\mathbf{Q}_\omega(\delta\omega,\delta\omega) + o(\|\delta\omega\|^2)$, where $\mathbf{Q}_\omega$ is the BKM/QFIM quadratic form. This is the information-geometric structure used in Definitions 2.2 and 2.4 and throughout Section 3. The broader trilogy interpretation also assumes Paper I's shell interface (Amb), dynamical covariance (C), and scale-faithfulness (SFa). These give conditional algebraic structure; modular flow is algebraic time, not a derived phenomenal-time variable.

2. **Small-ball expansion with $O(R^{d+3})$ remainder** (Paper II[2]). In the CHM ball framework for conformal nets, the modular energy of a small ball of radius R admits the expansion $\delta\langle K_{B,u}\rangle = \frac{2\pi V_d}{d+2} R^{d+1}\delta\langle T_{\mu\nu}\rangle u^\mu u^\nu + O(R^{d+3})$. The leading coefficient extracts the local stress tensor, while the higher-order remainder is uncontrolled. Paper II supplies neither an infinite coefficient tower nor a Fisher map; Section 5 uses only a separate exact $d = 1$ moment calculation.

3. **Self-reference motivation** (Paper G[3]). Paper G discusses regress and collapse scenarios but does not prove that a self-referential fixed point generically fails to exist. Paper III uses it as motivation only and does not quantify it.

4. **Ego-rigidity prior** (T-DOME Paper III[4]). The ego-rigidity prior $\mathcal{I}_{\text{ego}}$ is the Fisher information of the observer's prior distribution over its own self-model parameters: $\mathcal{I}_{\text{ego}} := \int(\partial_\sigma \log \pi_{\text{ego}}(\sigma))^2 \pi_{\text{ego}}(\sigma)\, d\sigma$. It may enter a separately specified Bayesian likelihood model. It does not lower-bound reconstruction loss without such an observation model, and no theorem in this paper uses it to do so.

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